

TAMIL NADU STATE BOARD - STANDARD 10 MATHEMATICS

CHAPTER 3: ALGEBRA

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 10 English Medium
Subject:	Mathematics
Chapter Name:	Algebra

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Linear Equation	An algebraic equation of first degree, which forms a straight line when plotted on a graph.
GCD of Polynomials	The Greatest Common Divisor of two or more polynomials is the polynomial of highest degree that divides each without remainder.
Quadratic Equation	A second-degree polynomial equation of the form $ax^2 + bx + c = 0$, where a is not zero.
Matrix	A rectangular array of numbers or expressions arranged in horizontal rows and vertical columns.

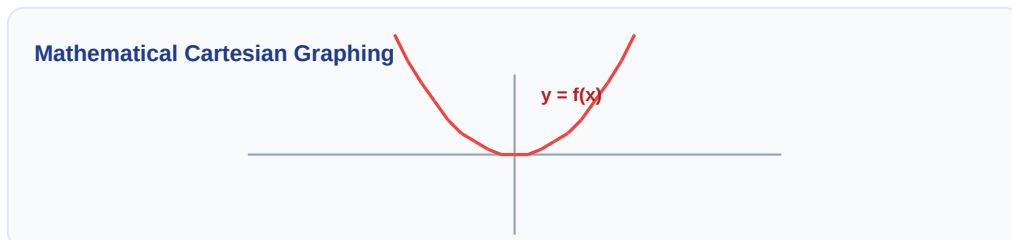
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Quadratic Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Discriminant: $\Delta = b^2 - 4ac$
- Nature of Roots: $\Delta > 0$ (Real & Unequal), $\Delta = 0$ (Real & Equal), $\Delta < 0$ (No Real Roots)
- Sum of Roots: $\alpha + \beta = -b/a$ | Product of Roots: $\alpha * \beta = c/a$
- Matrix Multiplication: Row-by-Column dot product

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Solve the quadratic equation $x^2 - 5x + 6 = 0$.

Step-by-step Solution:

Factorise the equation: $(x - 2)(x - 3) = 0$.

This gives two roots: $x = 2$ and $x = 3$.

Alternatively, using formula: $a=1$, $b=-5$, $c=6$. Discriminant $= 25 - 24 = 1$.

Roots $= (5 \pm 1)/2 \Rightarrow x = 3$ or $x = 2$.

Example 2: If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$, find AB .

Step-by-step Solution:

Row 1 x Col 1: $1 \cdot 2 + 2 \cdot 1 = 4$. Row 1 x Col 2: $1 \cdot 0 + 2 \cdot 2 = 4$.

Row 2 x Col 1: $3 \cdot 2 + 4 \cdot 1 = 10$. Row 2 x Col 2: $3 \cdot 0 + 4 \cdot 2 = 8$.

Result $AB = \begin{bmatrix} 4 & 4 \\ 10 & 8 \end{bmatrix}$.

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Find the values of k for which the quadratic equation $kx^2 - 10x + 5 = 0$ has real and equal roots.

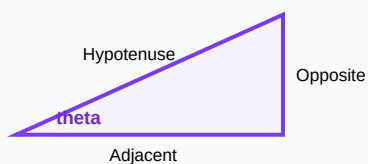
Analytical Solution Strategy:

For real and equal roots, the discriminant Δ must be zero.

$$\Delta = b^2 - 4ac = (-10)^2 - 4(k)(5) = 0.$$

$$100 - 20k = 0 \Rightarrow 20k = 100 \Rightarrow k = 5.$$

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Find the GCD of polynomials $x^3 - x^2$ and $x^2 - x$.

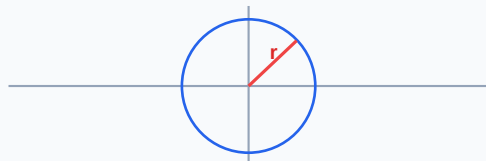
Q2 (2-Mark): Determine the nature of roots for $2x^2 - 3x + 4 = 0$.

Q3 (2-Mark): If matrix A has order 3×2 and B has order 2×4 , find the order of AB.

Q4 (5-Mark): Factorise the quadratic expression: $2x^2 + 7x + 6$.

Q5 (5-Mark): Solve the system: $x + y = 5$, $2x - y = 4$.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Image Processing

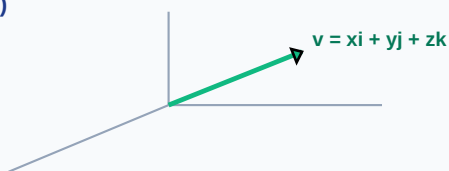
Digital images are represented as massive matrices of pixel values; filters apply matrix transformations to edit color.

Application Case: Aerodynamics

Quadratic equations model projectile trajectory paths of rockets and ballistics, calculating maximum heights.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

